# Closing the Numeracy Gap: Using Deliberation-Based Interventions to Equalize Decision Quality

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## Author Note

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All project materials, including experimental tasks, datasets, and analysis code are available in the OSF repository. Preregistrations for Study 1, Study 2, and Study 3 are available at their respective OSF links. The work was supported by the National Science Centre, Poland (Grant No. 2023/49/N/HS6/03244).

We have used AI tools (such as ChatGPT and Gemini) to improve the overall language, refine grammar, and do basic proofreading. We have no conflict of interest to declare.

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**Abstract**

Numeracy, the ability to process and understand probabilistic information, is a primary predictor of financial literacy, even after controlling for demographics, cognitive, and emotional factors. According to Fuzzy Trace Theory, highly numerate individuals achieve superior outcomes by engaging in deliberative processes that produce precise verbatim representations of a problem. In contrast, less numerate individuals often rely on simplified, less accurate gist representations. Across one simulation and three empirical studies, we evaluated a deliberation-based intervention designed to narrow this representational gap and support less numerate individuals in making decisions that align with normative standards. To do so, we developed a similarity measure, $\delta(n)$, that quantifies how closely the sample distribution generated during exploration approximates the true distribution of a probabilistic outcome. This measure enabled us to construct differential representations across three levels of deliberation and examine their impact on decision accuracy. Using a within-subjects design across the three empirical studies ($N$ = 600), we found a robust positive effect of deliberation on decision accuracy. Importantly, less numerate individuals benefited disproportionately from increased deliberation, showing a steeper improvement in accuracy relative to their more numerate counterparts. These findings highlight the utility of the similarity measure for facilitating the precise verbatim representations necessary for less numerate individuals to improve decision quality. The deliberation-based intervention consistently reduced the performance gap between more and less numerate individuals by improving the decision accuracy of less numerate individuals.

Keywords: intervention; numeracy; Fuzzy Trace Theory; deliberation; representation

# Closing the Numeracy Gap: Using Deliberation-Based Interventions to Equalize Decision Quality

Imagine you are considering a medical screening after being informed by your doctor that it reduces your risk of developing a serious disease by 25%. Many people assume this means the screening offers a large benefit and decide it is clearly worth doing. However, to make an informed decision, one must understand what that 25% reduction actually means. If your original risk is 4%, then a 25% relative risk reduction lowers it to 3%; an absolute reduction of only 1 percentage point. Without understanding such numerical information, people may overestimate the benefit of medical interventions and make poorly informed decisions. Navigating such choices requires high numeracy competency (the ability to understand probabilistic and numerical concepts). Over the past 25 years, research on numeracy has consistently indicated that highly numerate individuals are more risk literate, better understand the statistical regularities of the decision environment, and can leverage this understanding to accumulate greater wealth (Garcia-Retamero et al., 2019; Rudd et al., 1999; Schwartz et al., 1997).

While modern society is saturated with numerical data, surveys reveal that one-third of respondents in developed countries fail to answer basic probability questions (OECD, 2019). Evidence indicates that highly numerate individuals tend to make better decisions partly because they devote more time to each problem and produce choices that align with expected-value maximization (Mondal, 2021; Mondal & Traczyk, 2023). Fuzzy Trace Theory (FTT) proposes that people encode multiple levels of representation (categorical, ordinal, and verbatim) roughly in parallel, but individuals differ in the weight they assign to each type of representation (Broniatowski & Reyna, 2018). In other words, people with different levels of numeracy may vary in both their ability and their tendency to extract or rely on these representations during decision-making, leading to systematic differences in decision outcomes across numeracy groups (Brust-Renck & Reyna, 2023; Peters et al., 2019). In the present research, we evaluate a deliberation-based intervention intended to narrow this representational gap and help less

numerate individuals make better decisions[1].

## Numeracy & Deliberation

Evidence from individual difference research consistently identifies numeracy as the most robust predictor of financial literacy, even after controlling for demographics (e.g., age, gender, education, income), cognitive, and emotional factors (such as intelligence, cognitive reflection, self-efficacy, and financial anxiety; Skagerlund et al., 2018). Highly numerate individuals exhibit greater sensitivity to changes in expected value (EV), resulting in more EV-maximizing choices. Traczyk, Sobkow, et al. (2018) showed that highly numerate individuals are more flexible in the face of changing task demands and make more adaptive choices (shifting between optimal and suboptimal strategies) more readily than their less numerate counterparts. Extending this work, Mondal et al. (2025) found that although highly numerate individuals typically rely on deliberative, normative strategies in time-unconstrained settings, they adaptively switch to faster, recurring suboptimal strategies in time-constrained settings. This flexibility reflects a more accurate metacognitive understanding of task structure and allows them to deploy deliberative strategies when time permits and efficient heuristics when it does not. Together, these findings suggest that highly numerate individuals do not make EV-consistent choices because they only know one strategy; rather, they make superior decisions because they understand when and how to allocate cognitive resources to achieve the best outcomes (Mondal & Traczyk, 2025).

Deliberation, often used as a proxy for rational thinking, is a crucial metacognitive mediator of performance differences between high- and low-numeracy individuals (Baron, 2008). Prior work consistently indicates that highly numerate individuals spend more time on problems than less numerate individuals and, in turn, achieve superior choice performance (Mondal, 2021; Mondal & Traczyk, 2023; Traczyk, Sobkow, et al., 2018). This gap is evident across diverse cognitive domains in which deliberation is strongly associated with, or directly contributes to, improvements in performance (Bago et al., 2020; Garcia-Retamero et al., 2016; Hertzog &

[1] In this manuscript, “better” or “superior” decision making refers to selecting the option with the higher expected value.

Robinson, 2005).

Ghazal et al. (2014) conducted a large study with highly educated participants (of whom 30–50% held graduate degrees) to examine the metacognitive processes underlying superior decision-making across gain, loss, and mixed financial domains. Their results reaffirmed the relationship between numeracy, deliberation, and decision performance (Baron, 2008; Hertzog & Robinson, 2005). As 61% of participants were highly numerate, the authors argued that the observed performance differences were unlikely to be driven by raw mathematical skill. Instead, deliberation served as the primary mediator explaining the observed performance difference across all three conditions. They concluded that high numeracy alone does not guarantee superior judgment; rather, metacognitive processes such as deliberation are essential for making optimal choices.

The positive effect of deliberation on choice performance is also evident in the Decision from Experience (DfE) framework, where the number of samples drawn (pre-decision search) serves as a proxy for deliberation (Wulff et al., 2018). Prior work indicates that highly numerate individuals sample more information and switch less frequently between alternatives, thereby forming more accurate representations of choice problem, leading them to make more normatively superior choices (Ashby, 2017; Traczyk, Lenda, et al., 2018). Mondal et al. (2025) found that, in resource-constrained environments, highly numerate decision makers engaged in more dynamic and flexible information sampling and employed discovery-driven search strategies, predominantly focusing on reducing uncertainty and uncovering unobserved outcomes (Spektor & Wulff, 2024). Notably, the effect of deliberation was dissociated when all participants were required to sample the same amount of information. Traczyk et al. (2020) demonstrated a causal link between deliberation and decision performance by showing that the performance gap disappeared once sampling was equalized across individuals.

Collectively, these findings point to a key mechanism distinguishing skilled from less skilled decision makers. Despite higher numerical competence, highly numerate individuals appear to develop a more accurate metacognitive understanding of task structure and the choice

environment, which supports superior decisions.

## Theoretical Consideration

Among theoretical accounts of the psychological mechanisms underlying numeracy, Fuzzy Trace Theory (FTT) offers particularly broad and testable predictions about performance differences between high- and low-numeracy individuals. FTT argues that individuals encode three distinct memory representations of the same numerical information in parallel: categorical gist, ordinal gist, and verbatim representations (Reyna & Brust-Renck, 2020). To illustrate how these representations operate, consider the following decision problem:

*A) 50% chance of winning 200 €, otherwise 0 €* *vs.* *B) win 90 €* for sure

At the categorical gist level, Option A can be represented as a choice in which one might win something or nothing, whereas Option B is a sure win. Because people prefer the simplest gist, many choose the certain but suboptimal Option B over the optimal but uncertain Option A. When decisions require more nuance, individuals can shift to an ordinal gist representation by comparing relative magnitudes without exact calculations. In this view, Option A offers a larger reward with a lower probability, whereas Option B offers a smaller reward with a higher probability. This allows decision makers to weigh trade-offs without computing exact values. Finally, when ordinal gist representations are insufficient to discriminate between options, a precise verbatim representation becomes necessary. This involves energy-intensive calculations of exact quantities, such as multiplying probabilities and outcomes to compute EV for each alternative (Broniatowski & Reyna, 2018; Reyna & Brainerd, 1995).

The loss-bet paradigm offers a counterintuitive yet informative illustration of how representational processes differ across different levels of numeric competencies. Decision makers, especially highly numerate decision makers, rate the attractiveness of a bet with a small loss (7/36 chance to win $9 and 29/36 chance to lose 5 cents) substantially higher than the same bet without any loss (7/36 chance to win $9 and 29/36 chance to win $0; Slovic et al., 2004). FTT argues that less numerate individuals rely on categorical gist, simplifying a win-only gamble to “a

gain," whereas highly numerate individuals use ordinal representations for affective mapping. This higher precision predicts that the contrast between winning $9 and losing 5 cents is greater than the contrast between winning $9 and winning nothing because losing is ordinally more distinct than winning (Peters et al., 2019; Reyna & Brust-Renck, 2020). Hess et al. (2011) found similar patterns when using pictographs to communicate medical screening information. Less numerate individuals often do not rely on exact numbers; instead, they derive meaning directly from the pictograph. Conversely, more numerate individuals often draw meaning from exact numbers (i.e., relying on verbatim representations). Similarly, Peters and Bjalkebring (2015) found that higher numeracy is positively associated with performing more explicit numerical operations (i.e., number comparisons and calculations) in decision tasks.

Theoretical propositions from FTT and evidence from process-tracing studies suggest that the performance gap between highly and less numerate individuals may stem from processing interference that disrupts the formation of salient gist representations and contributes to biases that are endemic to judgments of probability and risk (Reyna & Brainerd, 2008; Reyna & Farley, 2006). This perspective allows us to test whether the gap in decision accuracy across numeracy levels is fixed or can be reduced. Prior research proposes that less numerate individuals struggle to process number-based EV trade-offs, suggesting a stable limitation in computational ability (e.g., Ashcraft & Kirk, 2001; Cokely & Kelley, 2009; Stanovich & West, 2000, 2008). If this account is correct, providing additional time or information for deliberation should yield little improvement. In contrast, FTT argues that less numerate individuals are capable of processing such information but may default to simplified representations or experience interference that disrupts the formation of precise gist or verbatim representations.

By using our similarity measure (described in the following section) to guide exploration, we can test these ideas. If less numerate individuals show a steeper improvement in accuracy with increased deliberation, this would suggest that the performance gap is not fixed and can be reduced. Importantly, because we cannot directly observe what specific information (such as exact numbers or simplified patterns) participants extract during the viewing stage, we focus on

behavioral outcomes. Demonstrating the reliable effect of structured exploration on improving EV consistency helps clarify how systematic decision environments can support less numerate decision makers, offering a modest but meaningful step toward evaluating these theoretical accounts.

## Overview of the Present Study

Despite these theoretical predictions, a dearth of research has tested deliberation as a determinant of superior decision making. Over the last decade, most work has prioritized improving risk communication (e.g., graphical presentation formats, absolute vs. relative risk, and neutral framing) to improve risk perception and foster better choices (Gigerenzer et al., 2007; Lusardi et al., 2017). However, such interventions often lack generalizability across contexts. Interventions that target metacognitive processes may offer broader applicability. The current study addresses this gap by testing whether less numerate individuals' performance improves when they are mandated or motivated to spend more time on each problem. Across one simulation and three empirical studies, we implement a deliberation-based intervention designed to help decision makers form more accurate gist representations and, when needed, engage in time- and effort-intensive verbatim processing to make superior decisions, thereby reducing the performance gap between less and more numerate individuals.

## Development of Similarity Measure

We ran a simulation study to develop the similarity measure, $\delta(n)$, that yields precise predictions about how many samples (a proxy for deliberation) are needed to form a detailed, accurate verbatim representation of choice problems. We used a Decision from Experience (DfE) format in which participants learned outcome probabilities through exploration: they repeatedly clicked a box representing each gamble, and each click drew an outcome from that option's underlying probability distribution.

The $\delta(n)$ metric serves as a criterion for the accuracy of a participant's understanding of a gamble's EV. Specifically, it captures how close the sample mean is to the true EV (in percentage terms). As the sample size increases, the difference between successive means, $\mu(n)$ and

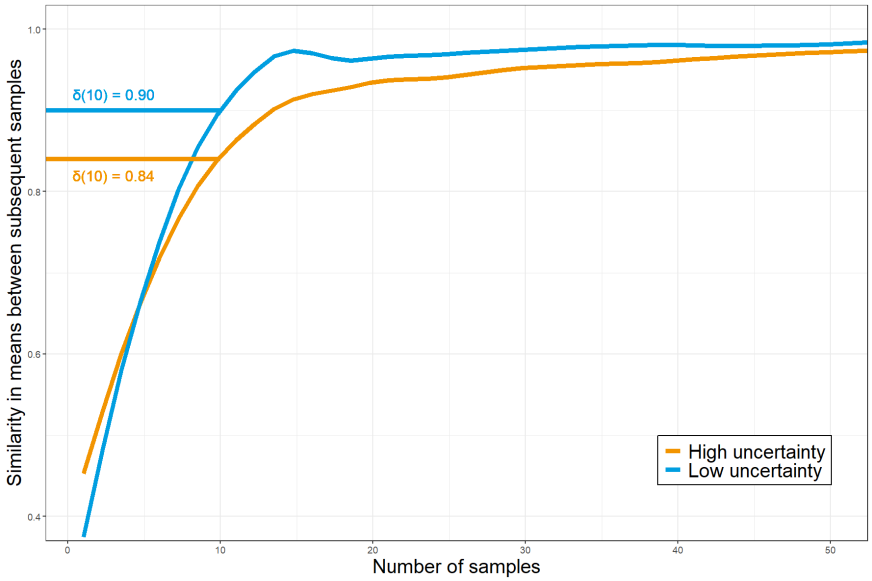
$\mu(n-1)$, diminishes.

$$\delta(n) = \begin{cases} 1 - \frac{|\mu(n)-\mu(n-1)|}{\mu(n-1)} & \text{if } \mu(n-1) > 0 \\ 0 & \text{if } \mu(n-1) = 0 \;\&\; \mu(n) > 0 \\ 1 & \text{if } \mu(n-1) = 0 \;\&\; \mu(n) = 0 \end{cases} \quad (1)$$

Here, $\delta(n)$ denotes the similarity measure for a given gamble; $\mu(n)$ and $\mu(n-1)$ denote the sample means after $n$ and $n-1$ draws, respectively.

**Figure 1**

*Similarity measure ($\delta(n)$). The measure quantifies similarity between the sample distribution generated during exploration and the true distribution by tracking changes in successive sample means (percentage change) as a function of sample size. After ten samples, the sample mean is closer to the true EV in a low-uncertainty environment (i.e., $\delta(10) = 0.90$) than in a high-uncertainty environment (i.e., $\delta(10) = 0.84$), indicating that higher uncertainty requires more exploration to achieve comparable veridicality.*




From an FTT perspective, increased sampling yields a more veridical understanding of the choice problem (i.e., more precise gist representations and/or more detailed verbatim representations; see Figure 1). Here, $\delta(n)$ provides a problem-specific criterion for determining

how much exploration is required. Rather than using a single fixed duration or sample size across all problems, we used $\delta(n)$ to define three deliberation levels: High (90% similarity), Moderate (80% similarity), and Low (70% similarity). This approach enabled us to control how many times participants sampled each option while also monitoring, for each choice problem, how closely the experienced outcome distribution approximated the true distribution. For example, problems with higher uncertainty (e.g., an outcome probability of 30%) required more sampling than lower-uncertainty problems (e.g., an outcome probability of 60%) to reach 90% veridicality (15 vs. 10 samples; see Figure 1). This similarity measure is therefore useful for studying metacognitive decision processes while controlling participants' representations throughout the intervention.

## General Method

### Participants

A total of 600 participants (53.37% male, $M_{age}$ = 43.03, $SD_{age}$ = 13.66) were recruited via Prolific and completed one of three online studies ($N_{Study1}$ = 192; $N_{Study2}$ = 123; $N_{Study3}$ = 285). Participants received an hourly base payment of £6.60, along with performance-based compensation of £1 for every 100 points earned on the decision task. Each study lasted approximately 20–30 minutes. Informed consent was obtained from all participants, and the study protocol was approved by the Ethics Committee at SWPS University (10/E/05/2025). Each participant was permitted to take part in only one study.

We applied consistent exclusion criteria across all empirical studies. For each study, we analyzed data only from participants who completed all tasks. We also removed duplicate submissions from the same participant ID and excluded participants who never selected the EV-maximizing option. In Studies 2 and 3[2], we additionally excluded participants who selected the non-dominated option more than once across three attention-check problems (see Table 4 in the Appendix). Based on these criteria, no participants were excluded from Study 1, two were excluded from Study 2, and fifteen were excluded from Study 3. Most exclusions resulted from

[2] Study 1 did not include attention-check problems; therefore, no attention-check exclusions were applied.

participants withdrawing shortly after beginning the procedure or failing the attention checks.

**Tasks**

The general procedure was similar across all three studies. The primary differences were confined to the viewing phase, during which participants deliberated over each option. Specifically, Study 1 and Study 2 differed in their sampling strategies, whereas the distinction between Study 2 and Study 3 concerned how sampling outcomes were experienced (see Figure 7 in the Appendix). Prior to the choice task, participants were familiarized with the task mechanics and bonus-payment procedure through training trials. During training, they answered comprehension questions to ensure they understood the task requirements. To proceed to the main task, participants had to answer these questions correctly; they received immediate feedback and repeated the training until all responses were correct.

After training, participants completed the choice task, which consisted of a viewing phase and a decision phase. In each study, participants completed the same 18 choice problems, divided equally between nine gain-domain and nine loss-domain problems (see Table 3 in the Appendix). During the viewing phase, participants experienced outcomes from each option for a stipulated duration. This duration, as predicted by the similarity measure, varied by deliberation condition (High, Moderate, Low). Choice problems assigned to each condition were held constant across studies, and presentation order was randomized for each participant. The number of samples drawn from each gamble during the viewing phase served as a proxy for deliberation (e.g., a participant might sample a problem 10 times in the Moderate condition but only six times in the Low condition).

Following the viewing phase, participants entered the decision phase and indicated their preference. Immediately afterward, they reported their confidence by answering, “How confident are you that you made the right decision?” using a slider from 0 (“Not at all confident”) to 10 (“Extremely confident”). After submitting their confidence rating, participants received trial-by-trial outcome feedback. In addition to the choice task, we administered scales measuring individual differences. Task order was counterbalanced across participants. No other variables

were manipulated, and none of the studies employed blinding.

### Individual Measures

Across the three studies, participants completed measures of statistical numeracy. We assessed numeracy using the three-item numeracy scale developed by Schwartz et al. (1997) together with the four-item Berlin Numeracy Test (BNT; Cokely et al., 2012), providing a comprehensive yet efficient assessment. The Schwartz scale captures basic probabilistic understanding and fundamental numerical skills, whereas the BNT assesses more advanced competencies such as risk literacy and reasoning with probabilities. Combining these measures broadens coverage of the numeracy construct while minimizing participant burden and reducing potential ceiling or floor effects, thereby yielding a more reliable measure of statistical numeracy (Ghazal et al., 2014; Skagerlund et al., 2018; Sobkow et al., 2025). We also assessed participants' familiarity with the numeracy items by asking whether they had previously answered them in other studies. The order of presentation was consistent across all three studies. Scores ranged from 0 to 7, with higher scores indicating greater statistical numeracy.

### Transparency and Openness

We report how we determined our sample size, all data exclusions (if any), all manipulations, and all measures in the study, and the study follows JARS (Appelbaum et al., 2018). All project materials, including experimental tasks, datasets, and analysis code are available in the OSF repository. Preregistrations for Study 1, Study 2, and Study 3 are available at their respective OSF links. Data were analyzed using R version 4.5.2 (R Core Team, 2020) and the packages tidyverse 2.0.0, ggeffects 2.3.2, and patchwork 1.3.2.

## Study 1

The objective of Study 1 was to investigate the effectiveness of a deliberation-based intervention in improving decision quality among less numerate individuals. We predict that mean EV consistency in the High deliberation condition would be significantly higher than in the Moderate and Low deliberation conditions. We also hypothesize that performance improvements across the three deliberation conditions would be larger for less numerate than for more numerate

individuals. Finally, we plan to examine the effect of deliberation on decision confidence. Based on prior research, we hypothesize that confidence ratings of decision makers will improve significantly with increased deliberation on each choice problem; specifically, the mean confidence rating in the High deliberation condition will be significantly higher than in both the Moderate and Low deliberation conditions.

To examine these hypotheses, we used the Decision from Experience (DfE) paradigm (Hertwig et al., 2004), in which participants were presented with two boxes representing binary, two-outcome gambles with unknown payoff distributions. Participants were informed of the required number of samples and how many samples they had already drawn from each option. Each time a gamble was selected during the viewing phase, an outcome was randomly drawn from its distribution and displayed for 400 ms. Participants could freely choose which gamble to explore and when to switch between gambles. Once participants reached the sampling limit predicted by the similarity measure, they transitioned to the decision phase and indicated their preference by selecting one of the gambles. In addition, participants' total accumulated gain was continuously displayed on the bottom-left corner of the screen (see the OSF repository for a video illustration). We recorded participants' preferences, the number of decisions made, and the total samples drawn during exploration. We also calculated the switching ratio, defined as the proportion of actual switches between gambles relative to the total number of possible switches given the number of samples drawn.

In addition to the choice task, we measured participants' numeracy. In Study 1, we also administered the Verbal Cognitive Reflection Test (VCRT) to assess participants' tendency to override intuitive but incorrect responses in favor of more deliberative, analytically grounded reasoning. Unlike the original numerical version, the VCRT uses verbally framed items that reduce reliance on mathematical ability while still capturing reflective cognitive processing, making it useful for distinguishing cognitive reflection from objective numeracy (Sirota et al., 2021). Scores ranged from 0 to 7, with higher scores indicating greater cognitive reflection. We also assessed participants' familiarity with the VCRT items by asking whether they had

previously answered them in other studies. Finally, we measured fluid intelligence using four matrix reasoning items from the International Cognitive Ability Resource (ICAR; Condon & Revelle, 2014), a widely used open-access battery designed to measure general cognitive ability. Scores ranged from 0 to 4, with higher scores indicating greater fluid intelligence.

## Results and Discussion

The goal of Study 1 was to examine the effectiveness of a deliberation-based intervention for improving decision quality among less numerate individuals. Following our preregistered analysis plan[3], we first conducted a repeated-measures ANOVA (RMANOVA) to examine the effect of deliberation on EV consistency (a measure of decision accuracy).

We found, using a Huynh–Feldt correction, a significant positive effect of deliberation on decision accuracy ($F(1.96, 373.42) = 25.57$, $p < .001$, $\eta_p^2 = .12$). Bonferroni-corrected post hoc comparisons, as captured by Table 1, indicated a significant increment in decision accuracy between the High and Low deliberation conditions (*Mean Difference* = 0.13, *SE* = 0.02, $t = 7.30$, $p_{\text{Bonf}} < .001$, $d = .65$) and between the High and Moderate deliberation conditions (*Mean Difference* = 0.11, *SE* = 0.02, $t = 5.51$, $p_{\text{Bonf}} < .001$, $d = .53$). However, there was no significant difference between the Moderate and Low deliberation conditions (*Mean Difference* = 0.03, *SE* = 0.02, $t = 1.17$, $p_{\text{Bonf}} = .732$, $d = .12$).

**Table 1**

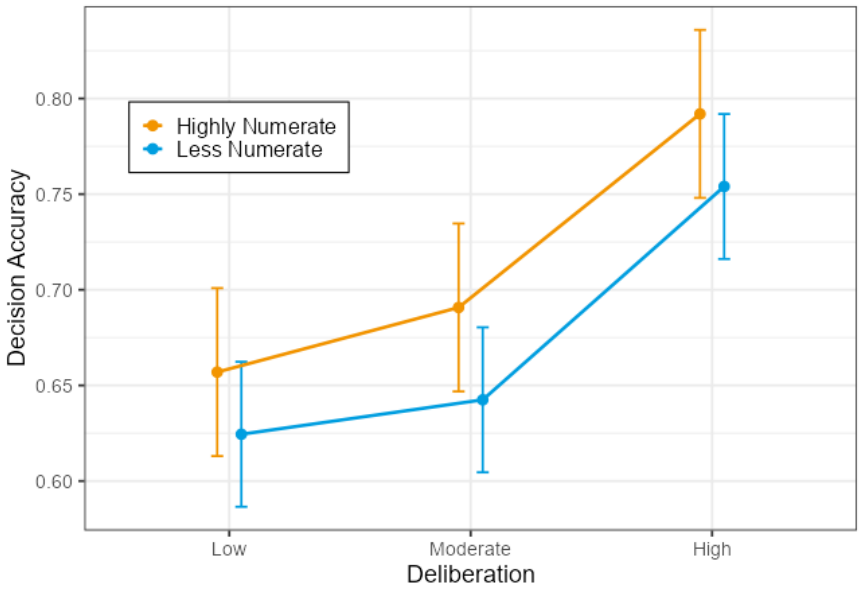
*Post hoc pairwise comparisons of decision accuracy across deliberation conditions in Study 1 (estimated marginal means).*

| Comparison | Mean Difference | SE | t | Cohen's d | $p_{\text{Bonf}}$ |
|---|---|---|---|---|---|
| High – Low | 0.132 | 0.018 | 7.292 | 0.650 | $< .001$ |
| High – Moderate | 0.107 | 0.019 | 5.512 | 0.528 | $< .001$ |
| Moderate – Low | 0.025 | 0.021 | 1.169 | 0.122 | .732 |

[3] Complete results of all analyses are reported in the Supplementary Materials document.

**Figure 2**

*Mean decision accuracy (EV consistency) by deliberation condition and numeracy group in Study 1. Error bars represent 95% confidence intervals.*




Next, in line with our preregistration, we conducted a linear trend analysis to examine whether the slope of performance improvement across the three deliberation conditions differed by numeracy. We categorized 192 participants into two numeracy groups ($N_{High}$ = 110; $N_{Low}$ = 82) using a median split (median = 4). Overall, consistent with prior research, highly numerate individuals made significantly more EV-consistent choices ($F$(1, 190) = 4.52, $p$ = .035, $\eta_p^2$ = .02) across the three deliberation conditions ($M_{High} = 0.79$, $SD_{High} = 0.17$; $M_{Moderate} = 0.69$, $SD_{Moderate} = 0.21$; $M_{Low} = 0.65$, $SD_{Low} = 0.20$) compared to less numerate individuals ($M_{High} = 0.75$, $SD_{High} = 0.20$; $M_{Moderate} = 0.64$, $SD_{Moderate} = 0.21$; $M_{Low} = 0.62$, $SD_{Low} = 0.21$). However, we found, using a Huynh–Feldt correction, no significant interaction between deliberation conditions and numeracy levels ($F$(1.96, 371.53) = 0.08, $p$ = .918, $\eta_p^2$ = .00). Yet the RMANOVA revealed a significant linear trend[4] for the interaction between numeracy and deliberation condition, $t$(190) = -7.52, $p$ < .001, $d$ = -0.69, indicating that the slope of

[4] The quadratic interaction trend did not reach statistical significance, $t$(190) = 1.85, $p$ = .065, $d$ = .18, nor did any higher-order comparisons ($ps$ > .05).

improvement in decision accuracy differed between the two numeracy groups. Based on the contrast coefficients, both groups showed significant improvement in decision accuracy; however, the negative contrast coefficient estimate of -0.14 ($SE$ = 0.02) indicates that less numerate participants improved more across the deliberation conditions than highly numerate participants (see Figure 2).

Finally, in accordance with our preregistered analysis plan, we assessed the effect of deliberation on participants' confidence ratings. We found, using a Huynh–Feldt correction, a significant effect of deliberation on decision confidence ($F(1.90, 362.35) = 3.31$, $p = .047$, $\eta_p^2 = .02$). However, the Bonferroni-corrected post hoc comparisons and the small effect size ($\eta_p^2 = .016$) suggest that this effect is not meaningful (see Figure 3A). The marginal mean difference in confidence between the High and Low deliberation conditions was negligible (*Mean Difference* = 0.00015, $SE$ = 0.01, $t$ = -0.002, $p_{\text{Bonf}} = 1.000$, $d$ = 0). Similarly, the difference between the High and Moderate deliberation conditions was not significant (*Mean Difference* = 0.17, $SE$ = 0.07, $t$ = 2.36, $p_{\text{Bonf}} = .058$, $d$ = .09).

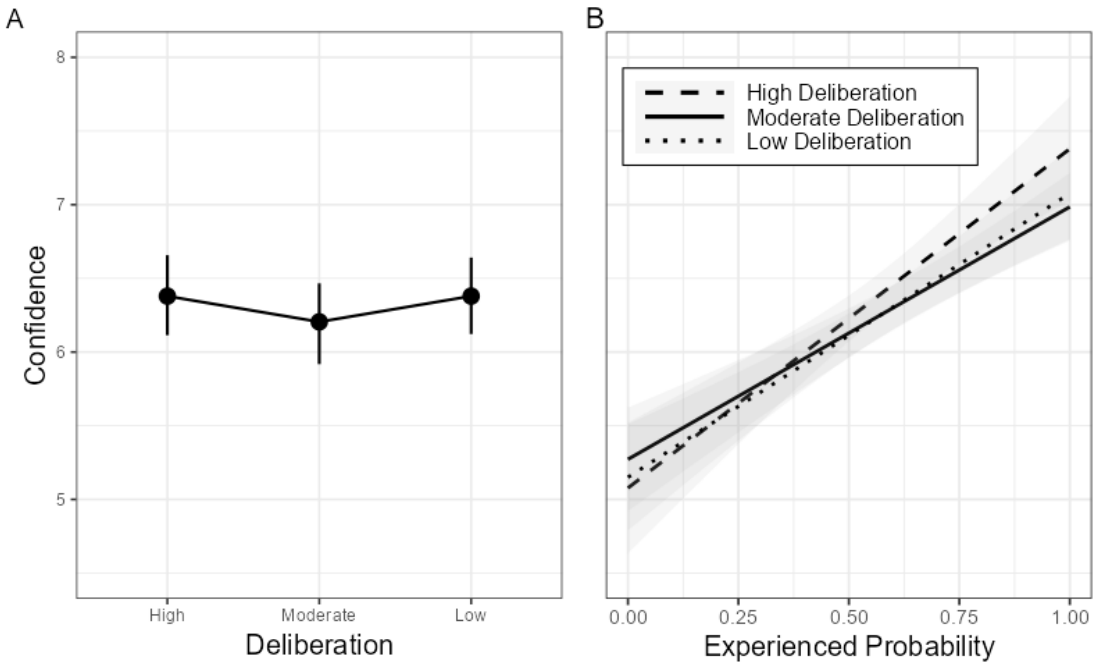
**Exploratory Analysis**

The goal of the exploratory analysis was to examine whether the positive effect of deliberation on decision accuracy persisted after controlling for individual differences in fluid intelligence (ICAR) and cognitive reflection (VCRT). The main effect of deliberation remained statistically significant (Huynh–Feldt corrected), $F(1.96, 369.73) = 5.89$, $p = .003$, $\eta_p^2 = .03$, indicating that the efficacy of deliberation-based intervention was independent of these cognitive abilities.

Next, we explored the relationship between deliberation and confidence. Prior work suggests that greater deliberation is positively associated with confidence ratings (Mata et al., 2013; Scherer et al., 2015). However, in this study, we found no evidence of an association between deliberation and confidence. We also observed at most a negligible relationship between sample size (the total number of times participants explored both options) and confidence ratings ($r$ = -.03, $p$ = .052). We then investigated whether experienced uncertainty associated with each

**Figure 3**

*Decision confidence as a function of deliberation and experienced probability in Study 1. Panel A shows mean confidence ratings across deliberation conditions (error bars = 95% bootstrapped confidence intervals). Panel B shows the relationship between experienced probability and confidence across deliberation conditions (High = dashed; Moderate = solid; Low = dotted; bands = 95% confidence intervals).*




problem influenced participants' confidence ratings. Specifically, we hypothesized that, in an uncertain decision environment, confidence reflects perceived certainty about obtaining potential rewards, independent of the cognitive or exploratory processes that produced that belief. Thus, if participants' experience led them to perceive the environment as highly uncertain, their confidence ratings should mirror that uncertainty regardless of how much time they spent exploring it. To test this hypothesis, we analyzed the relationship between confidence ratings and experienced probabilities, calculated as the selected option's experienced probability during the viewing stage. We found a significant positive relationship between experienced probabilities and confidence ratings ($r = .19$, $p < .001$). This relationship (Figure 3B) remained consistent even

after controlling for deliberation condition[5] ($\beta$ = 1.90, $t(3452) = 11.09$, $p < .001$). Moreover, deliberation condition had no significant main effect on confidence ratings (all *ps* > .15), indicating that confidence did not differ significantly between the High, Low ($\beta = -0.15$, $p$ = .155), or Moderate deliberation conditions ($\beta = -0.14$, $p$ = .189).

**Summary**

Study 1 investigated the effectiveness of a deliberation-based intervention for improving decision quality among less numerate individuals. The results supported our first and second preregistered hypotheses. Specifically, less numerate individuals showed markedly greater improvements in decision accuracy across the three deliberation conditions than individuals with higher numeracy. However, we found no support for the third hypothesis: mean confidence ratings did not differ significantly across deliberation conditions. Instead, confidence appeared to be driven by task-based certainty. In highly uncertain environments, extensive exploration (High deliberation condition) may still result in low confidence, whereas highly certain environments may elicit higher confidence even with minimal exploration (Low deliberation condition).

## Study 2

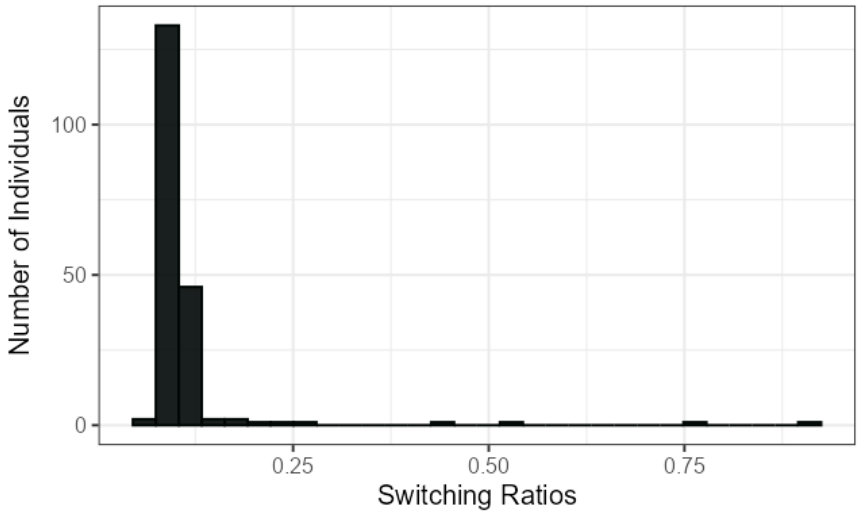
In Study 1, we showed that increased deliberation enabled participants, especially those with lower numeric ability, to form a more accurate representation of the choice problems, leading to better choices. However, as shown in Figure 4, this robust effect co-occurred with an unusually low switching ratio (*Median* = .10, *SD* = .09). The DfE literature suggests that search policies can be divided into two categories: comprehensive and piecewise (Hills & Hertwig, 2010). In a comprehensive strategy, people explore one option completely and then move to the other (i.e., switching very little), whereas in a piecewise strategy, people alternate between the options more frequently. Notably, studies consistently report a negative correlation between switching ratio and decision accuracy (Ashby, 2017; Mondal et al., 2025; Traczyk, Lenda, et al., 2018). This raises a critical question: Was the observed increase in decision accuracy a direct result of the deliberation

[5] This relationship remained robust even when we used chosen probabilities (i.e., the inherent probability of the chosen option) instead of experienced probabilities.

manipulation, or was it mediated by participants' tendency to use a comprehensive exploration strategy? We designed Study 2 to validate the direct causal link between deliberation and improved decision-making, independent of the behavioral artifact of reduced switching.

**Figure 4**

*Histogram of median switching ratios in Study 1.*



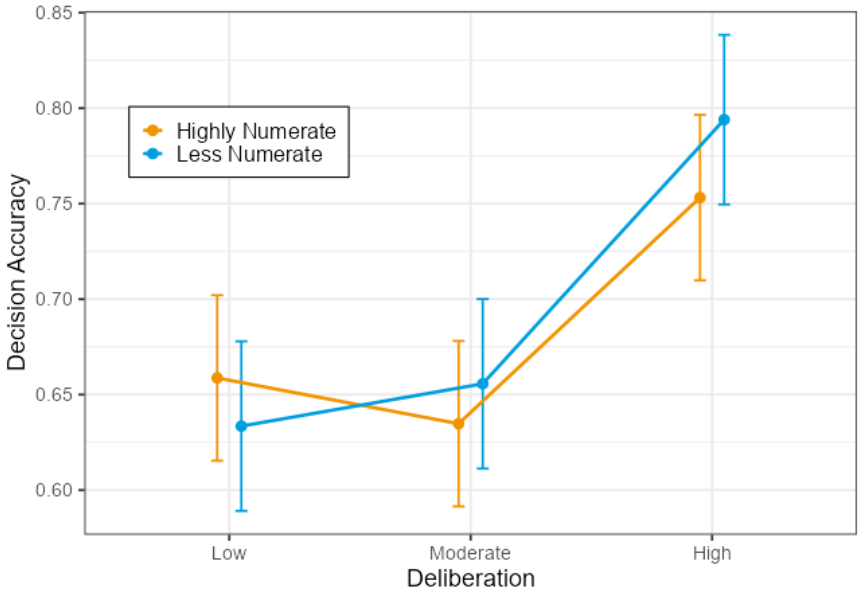

In Study 1, participants were told how many times they had to explore each option; however, they were free to choose which option to sample and in what order. Due to this freedom during the viewing phase in Study 1, participants predominantly used a comprehensive exploration strategy. In Study 2, we restricted this freedom by presenting a stream of outcomes such that participants were observers and did not voluntarily explore the options. In other words, during the viewing phase, we presented outcomes in a predetermined stream that mimicked a piecewise exploration strategy (without voluntary input), with participants observing outcomes from both options sequentially. This procedure ensured that the DfE task structure was comparable across the two studies (see the OSF repository for a video illustration). Consistent with Study 1, participants experienced outcomes during the viewing phase until the predicted sampling limit was met. Once the limit (predicted by the similarity measure) was met, participants transitioned to the decision phase and indicated their preference and confidence rating. In addition to the choice task, we measured each participant's objective numeracy.

## Results and Discussion

The goal of Study 2 was to examine the effectiveness of a deliberation-based intervention for improving decision quality among less numerate individuals after controlling for switching ratio. We successfully replicated the findings of Study 1 and found a significant positive effect of deliberation on decision accuracy ($F(2, 244) = 22.62$, $p < .001$, $\eta_p^2 = .16$). Bonferroni-corrected post hoc comparisons (Table 2) indicated a similar increment in decision accuracy between the High and Low deliberation conditions (*Mean Difference* = 0.13, *SE* = 0.02, $t = 6.32$, $p_{Bonf} < .001$, $d = .72$) and between the High and Moderate deliberation conditions (*Mean Difference* = 0.13, *SE* = 0.02, $t = 5.68$, $p_{Bonf} < .001$, $d = .73$). Consistent with Study 1, there was no significant difference between the Moderate and Low deliberation conditions (*Mean Difference* = -0.001, *SE* = 0.02, $t = -0.06$, $p_{Bonf} = 1.00$, $d = -.008$).

**Figure 5**

*Mean decision accuracy (EV consistency) by deliberation condition and numeracy group in Study 2. Error bars represent 95% confidence intervals.*




Next, we categorized participants into two numeracy groups ($N_{High} = 63$; $N_{Low} = 60$) using a median split and conducted a linear trend analysis. Unlike Study 1, we found no significant difference in EV consistency ($F(1, 121) = 0.41$, $p = .522$, $\eta_p^2 = .003$) across the three deliberation

**Table 2**

*Post hoc pairwise comparisons of decision accuracy across deliberation conditions in Study 2 (estimated marginal means).*

| Comparison | Mean Difference | SE | t | Cohen's d | $p_{\text{Bonf}}$ |
|---|---|---|---|---|---|
| High – Low | 0.128 | 0.023 | 5.688 | .733 | $< .001$ |
| High – Moderate | 0.127 | 0.020 | 6.323 | .724 | $< .001$ |
| Moderate – Low | −0.001 | 0.023 | −0.064 | −.008 | 1.000 |

conditions between highly numerate individuals ($M_{High} = 0.75$, $SD_{High} = 0.16$; $M_{Moderate} = 0.64$, $SD_{Moderate} = 0.20$; and $M_{Low} = 0.66$, $SD_{Low} = 0.17$) and less numerate individuals ($M_{High} = 0.79$, $SD_{High} = 0.16$; $M_{Moderate} = 0.66$, $SD_{Moderate} = 0.18$; and $M_{Low} = 0.63$, $SD_{Low} = 0.17$). Consistent with earlier results, we found no significant interaction between deliberation conditions and numeracy levels ($F(2, 242) = 1.20$, $p = .303$, $\eta_p^2 = .01$). However, we observed a significant linear interaction between numeracy and deliberation conditions, $t(121) = -5.82$, $p < .001$, $d = -0.67$, with a negative contrast coefficient estimate of -0.12 ($SE = 0.02$), indicating that less numerate participants improved significantly more across the three deliberation conditions than highly numerate participants (see Figure 5).

Finally, we assessed the effect of deliberation on participants' confidence ratings. In contrast to Study 1, deliberation had a significant effect on confidence ($F(2, 244) = 7.12$, $p < .001$, $\eta_p^2 = .06$), but in the opposite direction. Bonferroni-corrected post hoc comparisons indicated a significant decrement in confidence rating between the High and Low deliberation conditions (*Mean Difference* = -0.38, $SE = 0.11$, $t = -3.37$, $p_{\text{Bonf}} = .003$, $d = -0.25$) and between the Moderate and Low deliberation conditions (*Mean Difference* = -0.26, $SE = 0.10$, $t = -2.63$, $p_{\text{Bonf}} = .030$, $d = -0.18$). There was no significant difference between the High and Moderate deliberation conditions (*Mean Difference* = -0.12, $SE = 0.10$, $t = -1.21$, $p_{\text{Bonf}} = .684$, $d = -0.08$).

### Exploratory Analysis

The goal of this exploratory analysis was to validate the Study 1 findings where the level of experienced uncertainty influenced participants' confidence ratings. Consistent with Study 1, we found a significant positive relationship between experienced probabilities and confidence ratings ($r = .24$, $p < .001$). This relationship remained robust even after controlling for deliberation conditions [6] ($\beta = 2.20$, $t(2210) = 11.42$, $p < .001$). Moreover, deliberation condition had no significant main effect on confidence ratings (all $ps > .12$), indicating that confidence did not differ significantly between the High, Low ($\beta = 0.18$, $p = .137$), or Moderate conditions ($\beta = -0.19$, $p = .121$).

### Summary

Study 2 successfully validated, in accordance with our pre-registered hypotheses, the robust effect of deliberation-based intervention on decision accuracy. Consistent with Study 1, we found that an increment in deliberation enabled participants (especially less numerate participants) to create a more accurate representation of the choice problems, leading them to make superior choices. These findings suggest that the effect of deliberation is not contingent on a specific exploration strategy. In addition, we consistently found strong evidence that less numerate individuals showed greater improvements in decision accuracy across the three deliberation conditions than individuals with higher numeracy, regardless of whether they were using comprehensive or piecewise exploration strategies. Furthermore, we replicated our result from Study 1 which highlights that participants' experience of uncertainty moderates their decision confidence, regardless of the decision process involved.

## Study 3

Study 1 and Study 2 comprehensively demonstrated the effectiveness of a deliberation-based intervention for improving decision accuracy among less numerate

[6] Similar to Study 1, this relationship remained robust even if we used chosen probabilities (i.e., the inherent probability of the chosen option) instead of experienced probabilities.

individuals. Both studies used a DfE format to control the nature and quantity of information participants experienced during the viewing phase, thereby ensuring differential representations across the three deliberation conditions. As a result, one could argue that the effect of deliberation is inextricably tied to the DfE paradigm. Therefore, the goal of Study 3 was to disentangle the relationship between deliberation length and the DfE task paradigm. Specifically, we aimed to determine whether the observed improvement arises primarily from the amount of time participants spend on the task, the amount of information they acquire through exploration, or the degree of engagement (i.e., experiencing probabilistic outcomes) they exhibit while working through the problem.

To disentangle this relationship, we implemented a between-groups design comparing the Description-plus-Experience (DpE) and Decision-from-Description (DfD) task paradigms. In Study 1 and Study 2, participants built representations of each option through repeated exploration during the viewing phase, whereas in Study 3, participants in the DpE format received a description of each option (e.g., an *x*% chance of getting *y*) alongside a stream of outcomes representing exploration. Thus, participants had access to both the explicit description of probabilities and the ability to experience the probabilistic outcomes through exploration. Put simply, the primary difference between Study 2 and Study 3 was the introduction of descriptions for each option during the viewing phase. In the DfD format, participants observed only the description of the problem and did not experience the stream of numbers representing exploration (see the OSF repository for a video illustration). The current procedure builds upon the procedure implemented in Study 2 while introducing descriptions to eliminate the effect of exploration on decision accuracy across the three deliberation conditions. In other words, the introduction of descriptions in Study 3 enables us to examine whether the effect of deliberation on decision accuracy is moderated by exploration per se. We expect that regardless of the experimental condition (i.e., DpE or DfD), higher deliberation will lead to significant improvements in decision accuracy. In addition, we hypothesize that deliberation will yield significantly higher accuracy and a significantly greater magnitude of improvement in accuracy for participants in the DpE

condition compared to those in the DfD condition, due to the combined exposure to both descriptive and exploratory probabilistic task information.

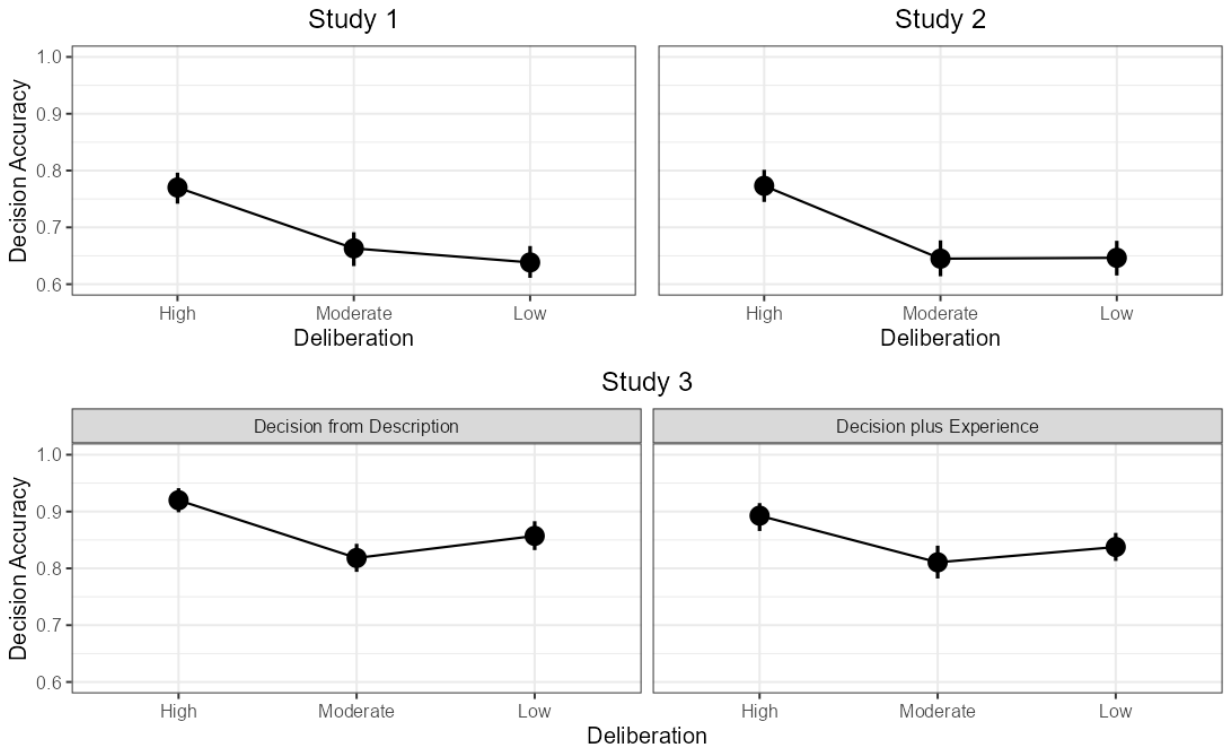
## Results and Discussion

To disentangle this relationship between experience and decision improvement across the three deliberation conditions, we performed an RMANOVA. The model included presentation format (DfD vs. DpE) as a between-subjects factor and deliberation levels (Low, Moderate, High) as a within-subjects factor. We found, using a Huynh-Feldt correction, a significant positive effect of deliberation on decision accuracy ($F$ (1.96, 555.27) = 40.76, $p < .001$, $\eta_p^2 = .13$). Bonferroni-corrected post hoc comparisons, consistent with results from Study 1 and Study 2, indicated that there was a significant increment in decision accuracy between the High and Low deliberation conditions (*Mean Difference* = 0.06, *SE* = 0.01, $t = 5.59$, $p_{\text{Bonf}} < .001$, $d = .39$) and between the High and Moderate deliberation conditions (*Mean Difference* = 0.10, *SE* = 0.01, $t = 9.71$, $p_{\text{Bonf}} < .001$, $d = .60$). In addition, we found a significant difference between the Moderate and Low deliberation conditions (*Mean Difference* = -0.03, *SE* = 0.01, $t = -3.03$, $p_{\text{Bonf}} = .008$, $d = .22$).

However, inconsistent with our preregistered hypothesis, we found no significant difference in decision accuracy between presentation formats ($F(1, 283) = 1.76$, $p = .186$, $\eta_p^2 = .006$). In addition, we found no interaction between deliberation level and presentation format ($F(1.96, 555.27) = 0.46$, $p = .630$, $\eta_p^2 = .002$). Next, we performed a trend analysis using polynomial contrasts to examine the interaction between presentation format (DfD vs. DpE) and deliberation condition. The analysis revealed a significant linear interaction, $t(283) = -5.60$, $p < .001$, *Estimate* = -0.063, indicating that the rate of change across deliberation conditions differed between the two formats. Consistent with our preregistered hypothesis, the linear increase was significantly steeper in the DpE condition than in the DfD condition. The analysis also yielded a significant quadratic interaction, $t(283) = 5.53$, $p < .001$, *Estimate* = 0.057, suggesting that curvature across the three deliberation conditions also varied by presentation format. A significant effect was also observed for Comparison 4, $t(283) = -4.29$, $p < .001$. No other polynomial

**Figure 6**

*Mean Decision Accuracy as a Function of deliberation conditions across Study 1, 2, and 3. Error bars represent 95% bootstrap confidence intervals.*




contrasts reached statistical significance (*ps* > .36).

**Exploratory Analysis**

In Study 1 and Study 2, we found no significant positive effect of deliberation on decision confidence. Both studies used a DfE paradigm in which participants had to experience probabilistic outcomes to form internal representations of each gamble during the viewing phase. As a result, we found that decision confidence is moderated by participants' perceived certainty about obtaining potential rewards and not by deliberation conditions. In other words, when participants experienced a gamble as highly uncertain yet attractive, their confidence ratings reflected that uncertainty, even when they spent substantial time exploring the gamble. In the current study, however, outcomes and their associated probabilities were explicitly displayed

during the viewing phase in both task presentation formats (DfD and DpE). Using a Huynh–Feldt correction, we observed a significant main effect of deliberation on decision confidence ($F(1.98, 560.14) = 14.48$, $p < .001$, $\eta_p^2 = .05$) with no significant interaction involving task presentation format ($F(1.98, 560.14) = 0.54$, $p = .580$, $\eta_p^2 = .002$). Bonferroni-corrected post hoc comparisons showed that confidence was significantly higher in the High deliberation condition than in both the Low (*Mean Difference* = 0.27, *SE* = 0.06, $t = 4.94$, $p_{\text{Bonf}} < .001$, $d = .18$) and Moderate conditions (*Mean Difference* = 0.23, *SE* = 0.05, $t = 4.47$, $p_{\text{Bonf}} < .001$, $d = .15$). The comparison between the Moderate and Low deliberation conditions was not significant (*Mean Difference* = 0.05, *SE* = 0.06, $t = 0.81$, $p_{\text{Bonf}} = 1.000$, $d = .03$).

**Summary**

Study 3 robustly replicated the positive effect of deliberation on decision accuracy. Regardless of presentation format, this finding reinforces the effectiveness of deliberation-based interventions in promoting more accurate choices. Although we did not observe a significant main effect of presentation format on accuracy, we did find, consistent with our preregistered hypothesis, that the improvement in accuracy across the three deliberation conditions was significantly greater in the DpE format than in the DfD format. Furthermore, in contrast to Study 1 and 2, we observed a significant positive relationship between deliberation condition and decision confidence. Specifically, mean confidence ratings in the High deliberation condition were significantly higher than those in both the Moderate and Low deliberation conditions.

## Overall Discussion

The primary goal of this research was to eliminate the representational differences between high- and low-numeracy decision makers by implementing a deliberation-based intervention designed to support less numerate individuals in making better decisions. Across the three studies, our results support the utility of the similarity measure in predicting the sufficient number of samples required, serving as a proxy for deliberation, to help less numerate individuals make superior decisions. Collectively, these results show a clear positive influence of deliberation on decision quality across three distinct experimental conditions.

Deliberation has long been theorized as a key determinant of superior decision making. Numerous studies across disciplines have manipulated deliberation to examine its impact on decision accuracy, ranging from economic choices and medical decision scenarios to fake-news comprehension, bias-reduction tasks, and even chess decision making (Fazio, 2020; Moxley et al., 2012; Rubin et al., 2019; Scherer et al., 2015; Shirasuna et al., 2025). Theoretical developments in the judgment and decision making field have also treated deliberation as a fundamental assumption. A prominent example is dual-process theory, which posits that deliberation is essential for improving decision quality, and that insufficient deliberation often leads to inferior outcomes due to the influence of cognitive biases (Kahneman, 2011). Across many versions of dual-process theory, researchers have consistently highlighted the difference between two types of cognitive processing: the rapid, automatic, and intuitive operations of System 1 and the slower, more analytical, and deliberative operations of System 2 (Chaiken & Trope, 1999; Kahneman, 2011; Sloman, 1996; Stanovich & West, 1999). However, contemporary understanding of dual-process theory argues that these two systems are not mutually exclusive categories; rather each system typically correlates highly with a specific set of cognitive processes (Evans & Stanovich, 2013). Current deliberation-based interventions build on this idea to examine the benefits of deliberation through instruction based methods, where participants are either told to make rapid, heuristic responses or urged to engage in more effortful, analytical reasoning. Such instructional methods are often quite coarse and produce mixed results: in some circumstances, deliberation leads to more accurate judgment (Evans & Curtis-Holmes, 2005; Fazio, 2020; Lawson et al., 2020), while in others, it has no effect (Rubin et al., 2019; Scherer et al., 2015).

The current study adopts a fundamentally different approach. Instead of using instructional intervention procedures, we developed the similarity measure that determines the appropriate deliberation period (based on the specific characteristics of the decision problem) necessary for decision makers, especially those with lower numerical proficiency, to create an accurate representation of the problem, enabling them to make good decisions. We speculate that the mixed results observed in the literature may not reflect an inherent inability of deliberation to

improve decision-making; rather the failure of instructional methods to assist decision makers in forming accurate representations. The consistent effect of deliberation on decision accuracy witnessed in the current study argues for a more systematic approach. We believe the similarity measure effectively provides the precision required to create salient, compelling, and accurate representations, thereby removing factors endemic to judgments of probability and risk that typically lead to inaccurate decisions.

The positive effect of deliberation is not simply restricted to improvements in decision accuracy; it also exerts a significant and positive influence on decision confidence. The standard view suggests that deliberation results in accurate decisions, which in turn influence confidence ratings (Mata et al., 2013; Scherer et al., 2015; Thompson, 2009). Across Study 1 and 2, we found no significant improvement in decision confidence as a function of deliberation, despite observing a positive effect on decision accuracy. Instead, we found that in an experimental environment where participants learn about outcomes and associated probabilities through exploration, decision confidence is largely dependent on the experienced uncertainty of the chosen option. However, in Study 3, where we implemented a Decision-from-Description (DfD) task procedure, we found, consistent with existing literature, an effect of deliberation on decision confidence.

This result introduces a new dimension to the relationship between deliberation and confidence, suggesting the existence of modality-specific valuation systems even in meta-cognition (Garcia et al., 2024). Standard models traditionally posit a central system wherein information and metacognitive processes, regardless of their origin, are translated into a common unit to facilitate comparison. In contrast, our findings indicate a meaningful divergence between metacognitive processes derived from direct experience (the DfE task paradigm) and those grounded in symbolic descriptions (the DfD task paradigm). This pattern suggests the presence of two modality-specific representational systems, each supported by distinct cognitive and potentially neural architectures. Future research should examine this possibility more systematically to establish stronger empirical support.

Taken together, our findings suggest that the performance of all decision makers improves

systematically with increased deliberation on each choice problem. Importantly, the linear trend analyses across our studies revealed a nuanced pattern regarding individual differences in numeracy. Even in contexts where we observed no overall mean differences in accuracy between the numeracy groups depending on a deliberation level, the rate of improvement (the slope) across the three deliberation levels was significantly steeper for less numerate individuals compared to highly numerate individuals. This indicates that while highly numerate decision makers maintain relatively stable accuracy across conditions, less numerate decision makers benefit disproportionately as deliberation increases. By highlighting this steeper slope of improvement rather than a simple main effect of numeracy, the trend analysis demonstrates performance convergence, suggesting that structured deliberation successfully reduces the performance gap between more and less numerate individuals by improving the decision accuracy of less numerate individuals.

## References

Appelbaum, M., Cooper, H., Kline, R. B., Mayo-Wilson, E., Nezu, A. M., & Rao, S. M. (2018). Journal article reporting standards for quantitative research in psychology: The apa publications and communications board task force report. *American Psychologist*, *73*(1), 3.

Ashby, N. J. (2017). Numeracy predicts preference consistency: Deliberative search heuristics increase choice consistency for choices from description and experience. *Judgment and Decision making*, *12*(2), 128–139.

Ashcraft, M. H., & Kirk, E. P. (2001). The relationships among working memory, math anxiety, and performance. *Journal of experimental psychology: General*, *130*(2), 224.

Bago, B., Rand, D. G., & Pennycook, G. (2020). Fake news, fast and slow: Deliberation reduces belief in false (but not true) news headlines. *Journal of experimental psychology: general*, *149*(8), 1608.

Baron, J. (2008). *Thinking and deciding* (4th ed.). Cambridge University Press.

Broniatowski, D. A., & Reyna, V. F. (2018). A formal model of fuzzy-trace theory: Variations on framing effects and the allais paradox. *Decision*, *5*(4), 205.

Brust-Renck, P. G., & Reyna, V. F. (2023). Individual differences in numerical representations of risk in health decision making: A fuzzy-trace theory approach. *Risk Analysis*, *43*(3), 548–557.

Chaiken, S., & Trope, Y. (1999). *Dual-process theories in social psychology*. Guilford Press.

Cokely, E. T., Galesic, M., Schulz, E., Ghazal, S., & Garcia-Retamero, R. (2012). Measuring risk literacy: The berlin numeracy test. *Judgment and Decision making*, *7*(1), 25–47.

Cokely, E. T., & Kelley, C. M. (2009). Cognitive abilities and superior decision making under risk: A protocol analysis and process model evaluation. *Judgment and Decision making*, *4*(1), 20–33.

Condon, D. M., & Revelle, W. (2014). The international cognitive ability resource: Development and initial validation of a public-domain measure. *Intelligence*, *43*, 52–64.

Evans, J. S. B., & Curtis-Holmes, J. (2005). Rapid responding increases belief bias: Evidence for the dual-process theory of reasoning. *Thinking & Reasoning*, *11*(4), 382–389.

Evans, J. S. B., & Stanovich, K. E. (2013). Dual-process theories of higher cognition: Advancing the debate. *Perspectives on psychological science*, *8*(3), 223–241.

Fazio, L. (2020). Pausing to consider why a headline is true or false can help reduce the sharing of false news. *Harvard Kennedy School Misinformation Review*, *1*(2).

Garcia, B., Lebreton, M., Bourgeois-Gironde, S., & Palminteri, S. (2024). The impassable gap between experiential and symbolic values. *Nature Human Behaviour*, *7*, 611–626.

Garcia-Retamero, R., Cokely, E. T., Wicki, B., & Joeris, A. (2016). Improving risk literacy in surgeons. *Patient education and counseling*, *99*(7), 1156–1161.

Garcia-Retamero, R., Sobkow, A., Petrova, D., Garrido, D., & Traczyk, J. (2019). Numeracy and risk literacy: What have we learned so far? *The Spanish journal of psychology*, *22*, E10.

Ghazal, S., Cokely, E. T., & Garcia-Retamero, R. (2014). Predicting biases in very highly educated samples: Numeracy and metacognition. *Judgment and Decision making*, *9*(1), 15–34.

Gigerenzer, G., Gaissmaier, W., Kurz-Milcke, E., Schwartz, L. M., & Woloshin, S. (2007). Helping doctors and patients make sense of health statistics. *Psychological science in the public interest*, *8*(2), 53–96.

Hertwig, R., Barron, G., Weber, E. U., & Erev, I. (2004). Decisions from experience and the effect of rare events in risky choice. *Psychological science*, *15*(8), 534–539.

Hertzog, C., & Robinson, A. E. (2005). Metacognition and Intelligence. In O. Wilhelm & R. W. Engle (Eds.), *Handbook of understanding and measuring intelligence* (pp. 101–123). Sage Publications, Inc.

Hess, R., Visschers, V. H., & Siegrist, M. (2011). Risk communication with pictographs: The role of numeracy and graph processing. *Judgment and Decision making*, *6*(3), 263–274.

Hills, T. T., & Hertwig, R. (2010). Information search in decisions from experience: Do our patterns of sampling foreshadow our decisions? *Psychological science*, *21*(12), 1787–1792.

Kahneman, D. (2011). *Fast and slow thinking*. Macmillan.

Lawson, M. A., Larrick, R. P., & Soll, J. B. (2020). Comparing fast thinking and slow thinking: The relative benefits of interventions, individual differences, and inferential rules. *Judgment and Decision making*, *15*(5), 660–684.

Lusardi, A., Samek, A., Kapteyn, A., Glinert, L., Hung, A., & Heinberg, A. (2017). Visual tools and narratives: New ways to improve financial literacy. *Journal of Pension Economics & Finance*, *16*(3), 297–323.

Mata, A., Ferreira, M. B., & Sherman, S. J. (2013). The metacognitive advantage of deliberative thinkers: A dual-process perspective on overconfidence. *Journal of personality and social psychology*, *105*(3), 353.

Mondal, S. (2021). Sensitivity of numerate individuals to large asymmetry in outcomes: A registered replication of traczyk et al.(2018). *Decyzje*, (35), 5–26.

Mondal, S., Lenda, D., & Traczyk, J. (2025). Recurring suboptimal choices result in superior decision making. *Decision*, *12*(1), 63.

Mondal, S., & Traczyk, J. (2023). Conditionality of adaptiveness: Investigating the relationship between numeracy and adaptive behavior. *Journal of Economic Psychology*, *96*, 102611.

Mondal, S., & Traczyk, J. (2025). Adaptive decision making in the wild: A case study of chess. *Thinking & Reasoning*, 1–21.

Moxley, J. H., Ericsson, K. A., Charness, N., & Krampe, R. T. (2012). The role of intuition and deliberative thinking in experts' superior tactical decision-making. *Cognition*, *124*(1), 72–78.

OECD. (2019). *Skills matter: Additional results from the survey of adult skills*.

Peters, E., & Bjalkebring, P. (2015). Multiple numeric competencies: When a number is not just a number. *Journal of personality and social psychology*, *108*(5), 802.

Peters, E., Fennema, M. G., & Tiede, K. E. (2019). The loss-bet paradox: Actuaries, accountants, and other numerate people rate numerically inferior gambles as superior. *Journal of Behavioral Decision Making*, *32*(1), 15–29.

Reyna, V. F., & Brainerd, C. J. (1995). Fuzzy-trace theory: An interim synthesis. *Learning and individual Differences*, *7*(1), 1–75.

Reyna, V. F., & Brainerd, C. J. (2008). Numeracy, ratio bias, and denominator neglect in judgments of risk and probability. *Learning and individual differences*, *18*(1), 89–107.

Reyna, V. F., & Brust-Renck, P. G. (2020). How representations of number and numeracy predict decision paradoxes: A fuzzy-trace theory approach. *Journal of Behavioral Decision Making*, *33*(5), 606–628.

Reyna, V. F., & Farley, F. (2006). Risk and rationality in adolescent decision making: Implications for theory, practice, and public policy. *Psychological science in the public interest*, *7*(1), 1–44.

Rubin, E. B., Buehler, A. E., Cooney, E., Gabler, N. B., Mante, A. A., & Halpern, S. D. (2019). Intuitive vs deliberative approaches to making decisions about life support: A randomized clinical trial. *JAMA network open*, *2*(1), e187851.

Rudd, R. E., Moeykens, B. A., & Colton, T. C. (1999). Health and literacy: A review of medical and public health literature. *Office of Educational Research and Improvement*.

Scherer, L. D., de Vries, M., Zikmund-Fisher, B. J., Witteman, H. O., & Fagerlin, A. (2015). Trust in deliberation: The consequences of deliberative decision strategies for medical decisions. *Health Psychology*, *34*(11), 1090.

Schwartz, L. M., Woloshin, S., Black, W. C., & Welch, H. G. (1997). The Role of Numeracy in Understanding the Benefit of Screening Mammography. *Annals of Internal Medicine*, *127*(11), 966–972.

Shirasuna, M., Kagawa, R., & Honda, H. (2025). Pause before action: Waiting short time as a simple and resource-rational boost. *Scientific Reports*, *15*(1), 4362.

Sirota, M., Dewberry, C., Juanchich, M., Valuš, L., & Marshall, A. C. (2021). Measuring cognitive reflection without maths: Development and validation of the verbal cognitive reflection test. *Journal of Behavioral Decision Making*, *34*(3), 322–343.

Skagerlund, K., Lind, T., Strömbäck, C., Tinghög, G., & Västfjäll, D. (2018). Financial literacy and the role of numeracy–how individuals' attitude and affinity with numbers influence financial literacy. *Journal of behavioral and experimental economics*, *74*, 18–25.

Sloman, S. A. (1996). The empirical case for two systems of reasoning. *Psychological bulletin*, *119*(1), 3.

Slovic, P., Finucane, M. L., Peters, E., & MacGregor, D. G. (2004). Risk as analysis and risk as feelings: Some thoughts about affect, reason, risk, and rationality. *Risk Analysis*, *24*(2), 311–322.

Sobkow, A., Jurewicz, K., Zaleskiewicz, T., Pronczuk, A., Mondal, S., Badecka, J., Bartkiewicz, U., Bobowiec, W., Bochniak, M., Chwalibok, K., et al. (2025). The role of numeracy in judgment and decision making: The replication of eleven effects across various numeracy scales. *Collabra: Psychology*, *11*(1), 128514.

Spektor, M. S., & Wulff, D. U. (2024). Predecisional information search adaptively reduces three types of uncertainty. *Proceedings of the National Academy of Sciences*, *121*(47), e2311714121.

Stanovich, K. E., & West, R. F. (1999). Discrepancies between normative and descriptive models of decision making and the understanding/acceptance principle. *Cognitive psychology*, *38*(3), 349–385.

Stanovich, K. E., & West, R. F. (2000). Advancing the rationality debate. *Behavioral and brain sciences*, *23*(5), 701–717.

Stanovich, K. E., & West, R. F. (2008). On the relative independence of thinking biases and cognitive ability. *Journal of personality and social psychology*, *94*(4), 672.

Thompson, V. A. (2009). Dual process theories: A metacognitive perspective. In J. S. B. T. Evans & K. Frankish (Eds.), *In two minds: Dual processes and beyond* (pp. 171–195). New York, NY: Oxford University Press.

Traczyk, J., Lenda, D., Serek, J., Fulawka, K., Tomczak, P., Strizyk, K., Polec, A., Zjawiony, P., & Sobkow, A. (2018). Does Fear Increase Search Effort in More Numerate People? An Experimental Study Investigating Information Acquisition in a Decision From Experience Task. *Frontiers in Psychology*, *9*.

Traczyk, J., Sobkow, A., Fulawka, K., Kus, J., Petrova, D., & Garcia-Retamero, R. (2018). Numerate decision makers don't use more effortful strategies unless it pays: A process tracing investigation of skilled and adaptive strategy selection in risky decision making. *Judgment and Decision Making*, *13*(4), 372–381.

Traczyk, J., Sobkow, A., Matukiewicz, A., Petrova, D., & Garcia-Retamero, R. (2020). The experience-based format of probability improves probability estimates: The moderating role of individual differences in numeracy. *International Journal of Psychology*, *55*(2), 273–281.

Wulff, D. U., Mergenthaler-Canseco, M., & Hertwig, R. (2018). A meta-analytic review of two modes of learning and the description-experience gap. *Psychological bulletin*, *144*(2), 140.

# Appendix

## Figure 7

*Procedural differences between the three studies. ICAR = International Cognitive Ability Resource; VCRT = Verbal Cognitive Reflection Test.*

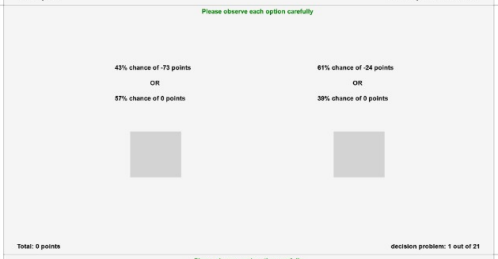
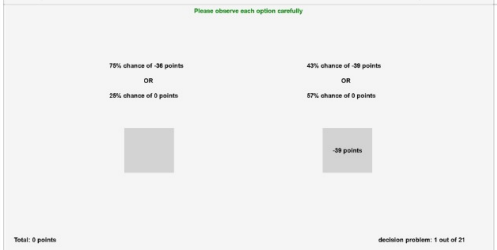
| Study | Screenshot | Presentation format | Measures | Number of Participants |
|---|---|---|---|---|
| **Study 1** |  | Decision from Experience (voluntary) | Choice problems, Numeracy Scale, VCRT, & ICAR | 192 |
| **Study 2** |  | Decision from Experience (Involuntary) | Choice problems & Numeracy Scale | 123 |
| **Study 3** |  | Decision from Description | Choice problems & Numeracy Scale | 149 |
| |  | Description plus Experience | Choice problems & Numeracy Scale | 136 |

**Table 3**

*Choice problems.*

| N | $o_{A1}$ | $p_{A1}$ | $o_{B1}$ | $p_{B1}$ | $o_{A2}$ | $p_{A2}$ | $o_{B2}$ | $p_{B2}$ | Deliberation | Exploration A | Exploration B | Total exploration |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | -72 | 0.41 | -26 | 0.46 | 0 | 0.59 | 0 | 0.54 | Moderate | 7 | 7 | 14 |
| 2 | -65 | 0.41 | -58 | 0.72 | 0 | 0.59 | 0 | 0.28 | High | 11 | 8 | 19 |
| 3 | 58 | 0.36 | 48 | 0.74 | 0 | 0.64 | 0 | 0.26 | Low | 4 | 3 | 7 |
| 4 | 29 | 0.72 | 13 | 0.74 | 0 | 0.28 | 0 | 0.26 | Low | 2 | 2 | 4 |
| 5 | 49 | 0.47 | 20 | 0.64 | 0 | 0.53 | 0 | 0.36 | Moderate | 7 | 5 | 12 |
| 6 | 43 | 0.7 | 52 | 0.37 | 0 | 0.3 | 0 | 0.63 | Low | 2 | 4 | 6 |
| 7 | 27 | 0.59 | 50 | 0.56 | 0 | 0.41 | 0 | 0.44 | High | 8 | 8 | 16 |
| 8 | -73 | 0.43 | -24 | 0.61 | 0 | 0.57 | 0 | 0.39 | Moderate | 5 | 4 | 9 |
| 9 | -50 | 0.3 | -36 | 0.71 | 0 | 0.7 | 0 | 0.29 | Low | 3 | 3 | 6 |
| 10 | 17 | 0.4 | 60 | 0.31 | 0 | 0.6 | 0 | 0.69 | Moderate | 7 | 4 | 11 |
| 11 | 15 | 0.61 | 42 | 0.57 | 0 | 0.39 | 0 | 0.43 | High | 10 | 9 | 19 |
| 12 | -54 | 0.3 | -44 | 0.71 | 0 | 0.7 | 0 | 0.29 | Low | 5 | 3 | 8 |
| 13 | -36 | 0.75 | -39 | 0.43 | 0 | 0.25 | 0 | 0.57 | Moderate | 3 | 6 | 9 |
| 14 | 70 | 0.46 | 61 | 0.71 | 0 | 0.54 | 0 | 0.29 | Moderate | 5 | 3 | 8 |
| 15 | -24 | 0.59 | -69 | 0.47 | 0 | 0.41 | 0 | 0.53 | High | 10 | 12 | 22 |
| 16 | -21 | 0.59 | -40 | 0.56 | 0 | 0.41 | 0 | 0.44 | High | 10 | 10 | 20 |
| 17 | -26 | 0.73 | -69 | 0.43 | 0 | 0.27 | 0 | 0.57 | Low | 3 | 4 | 7 |
| 18 | 18 | 0.73 | 37 | 0.72 | 0 | 0.27 | 0 | 0.28 | High | 7 | 6 | 13 |

**Table 4**

*Attention-check problems.*

| N | $o_{A1}$ | $p_{A1}$ | $o_{B1}$ | $p_{B1}$ | $o_{A2}$ | $p_{A2}$ | $o_{B2}$ | $p_{B2}$ | Deliberation | Exploration A | Exploration B | Total exploration |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | -45 | 0.70 | 0 | 1 | 0 | 0.30 | 0 | 0 | NA | 8 | 8 | 16 |
| 2 | 20 | 0.20 | 30 | 1 | 0 | 0.80 | 0 | 0 | NA | 8 | 8 | 16 |
| 3 | 0 | 1 | -30 | 0.50 | 0 | 0 | 0 | 0.50 | NA | 8 | 8 | 16 |